

Multi-Agent System: Theories and Applications

Lecture 7: Consensus of 1st- and 2nd-order Linear MASs

Nguyen Dinh Hoa

International Institute for Carbon-Neutral Energy Research (WPI-I²CNER)
and Institute of Mathematics for Industry (IMI)
Kyushu University

Outline

- 1 Consensus of Generic Linear MASs
- 2 Consensus of 1st-order Linear MASs
- 3 Consensus of 2nd-order Linear MASs
- 4 Harmonic Oscillator Network
- 5 Coupled RLC Circuits in Electrical and Electronic Engineering

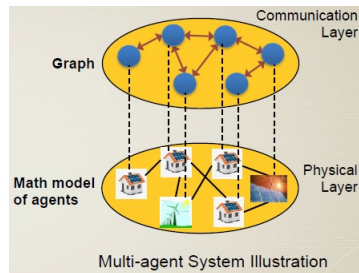
Revisit of Consensus Problem

Assumption: Connectedness of $\mathcal{G}$

$$\dot{x}(t) = (I_N \otimes A)x(t) + (I_N \otimes B)u(t)$$

$$y(t) = (I_N \otimes C)x(t)$$

$$A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{n \times m}, C \in \mathbb{R}^{p \times n}$$



Control Problem

Find $u(t)$ such that

$$\lim_{t \rightarrow t_f} x_i(t) = x^*, \forall i = 1, \dots, N \Leftrightarrow \lim_{t \rightarrow t_f} x(t) = \mathbf{1}_N \otimes x^*$$

for any initial conditions $x_i(0)$.

One Consensus Design for Generic Linear MASs

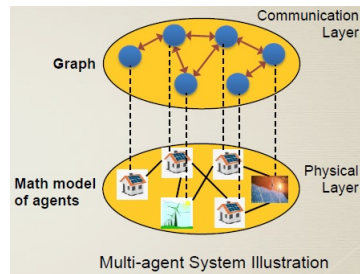
Assumption 1: $\mathcal{G}$ is undirected and connected

Assumption 2: (A, B) is controllable

Assumption 3: A has a single 0 eigenvalue and other eigenvalues in $\mathbb{C}_-$

$$\dot{x}(t) = (I_N \otimes A)x(t) + (I_N \otimes B)u(t)$$

$$A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{n \times m}, C \in \mathbb{R}^{p \times n}$$



Theorem 1

Consensus is achieved with

$$u(t) = -\mu(A - K)x(t), \quad K = RB^T P$$

for any $\mu > 0$, where $P \succ 0$ is the unique solution of the Riccati equation:

$$PA + A^T P - PBRB^T P + Q = 0$$

in which $Q \succeq 0$, $R \succ 0$, and $(Q^{1/2}, A)$ is observable.

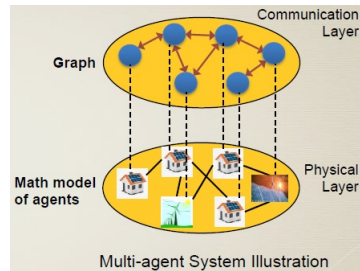
Proof: See: Dinh Hoa Nguyen, "Reduced-Order Distributed Consensus Controller Design via Edge Dynamics", IEEE Transactions on Automatic Control, vol. 62(1), pp. 475–480, 2017.

Consensus Design for 1st-Order Linear MASs

Assumption: $\mathcal{G}$ is undirected and connected

$$\dot{x}(t) = u(t)$$

$$x(t) \in \mathbb{R}^N, u(t) \in \mathbb{R}^N$$



Theorem 2

Consensus is achieved with

$$u(t) = -\mu \mathcal{L}x(t)$$

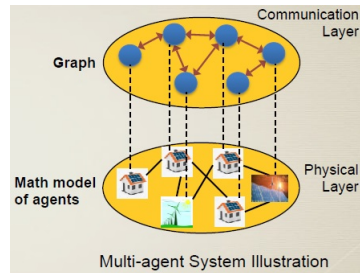
for any $\mu > 0$.

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How about with digraphs?

1st Proof

- Closed-loop dynamics:

$$\dot{x} = -\mu \mathcal{L}x \Rightarrow x(t) = e^{-\mu \mathcal{L}t} x(0)$$

- $\mathcal{G}$ is connected, hence $\mathcal{L}$ has a single eigenvalue 0 (Lemma 2), and $\mathcal{L} \succeq 0$.

- Diagonalization:

$$\mathcal{L} = U \text{diag}\{0, \lambda_2, \dots, \lambda_N\} U^T, \quad U \in \mathbb{R}^{N \times N}, \quad U U^T = I_N$$

$$\Rightarrow \lim_{t \rightarrow \infty} e^{-\mu \mathcal{L}t} = U e^{-\text{diag}\{0, \mu \lambda_2, \dots, \mu \lambda_N\} t} U^T = U \text{diag}\{1, 0, \dots, 0\} U^T$$

$$\Rightarrow \lim_{t \rightarrow \infty} x(t) = U \text{diag}\{1, 0, \dots, 0\} U^T x(0) = \frac{\mathbf{1}_N \mathbf{1}_N^T}{N} x(0)$$

$$\lim_{t \rightarrow \infty} x(t) = \frac{\mathbf{1}_N^T x(0)}{N} \mathbf{1}_N$$

average consensus

2nd Proof using Krasovskii-Lasalle Invariance Principle

- Choose a Lyapunov function candidate: $V(x) = x^T x$
- Compute derivative of $V(x)$: $\dot{V}(x) = -2\mu x^T Lx \leq 0 \rightarrow$ Lyapunov theorems cannot give the asymptotic stability
- Utilizing Krasovskii-Lasalle invariance principle, $x(t)$ converges to the set $\{x \in \mathbb{R}^n : \dot{V}(x) = 0\} \equiv \{x \in \ker(\mathcal{L})\}$
- Given $\mathcal{G}$ undirected and connected, $\ker(\mathcal{L}) \equiv \text{span}(\mathbf{1}_N)$, hence $x(t)$ comes to $\mathbf{1}_N x^*$ for some x^* (Q.E.D)

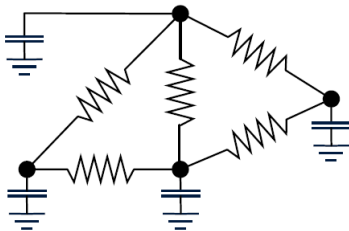
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How about digraphs?

RC Circuit as An Integrator MAS

Example of an RC circuit



- Dynamics of node k : $C_k \dot{v}_k = -i_k$, where i_k and v_k are injected current and voltage at node k .
- Using Kirchhoff's laws, dynamics of overall circuit: $\dot{\mathbf{v}} = -\mathcal{L}\mathbf{v}$, where $\mathcal{L}$ is Laplacian matrix of an undirected graph $\mathcal{G}$ representing the circuit
- Hence, $C \dot{\mathbf{v}} = -\mathcal{L}\mathbf{v} \Leftrightarrow \dot{\mathbf{v}} = - \underbrace{C^{-1}\mathcal{L}}_{\text{Laplacian matrix of a digraph}} \mathbf{v}$, where $C = \text{diag}(C_k)$

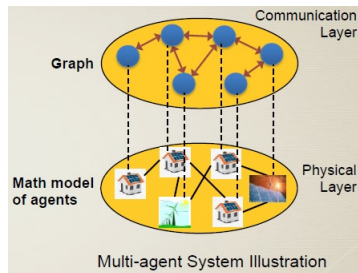
→ consensus of node voltages if $\mathcal{G}$ is connected

Consensus Design

Assumption: $\mathcal{G}$ is undirected and connected

$$\dot{x}(t) = (I_N \otimes A)x(t) + (I_N \otimes B)u(t)$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Theorem 3

Consensus is achieved with

$$u(t) = -(\mathcal{L} \otimes [k_1 \quad k_2])x(t)$$

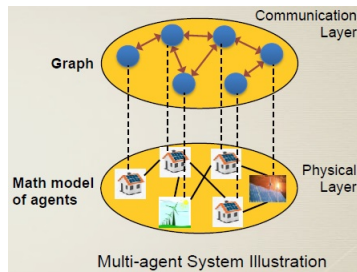
for any $k_1 > 0, k_2 > 0$.

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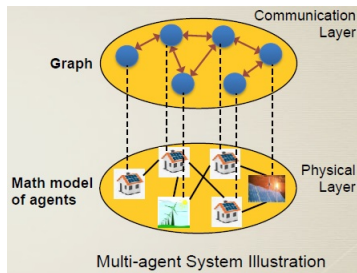
How about with digraphs?

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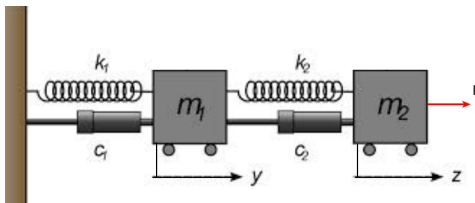
$$u(t) = -(\mathcal{L} \otimes [k_1 \quad k_2])x(t)$$

for any $k_1 > 0, k_2 > 0$.

How about with digraphs?

Can you prove using Krasovskii-Lasalle Invariance Principle?

Harmonic Oscillator in Mechanical Engineering



Model of Coupled Harmonic Oscillators

- Each oscillator:

$$\dot{x}_i = Ax_i + Bu_i, \quad A = \begin{bmatrix} 0 & 1 \\ -\alpha & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \alpha > 0$$

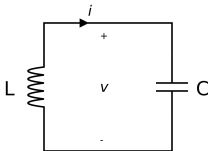
- Coupled oscillators:

$$\ddot{x} + \mathcal{L}\dot{x} + \alpha x = 0$$

c.f. consensus of double-integrator MASs

RLC Circuit as An Oscillator Network

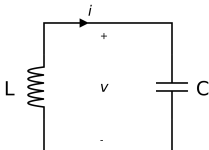
LC circuit as an oscillator



- RLC oscillators appear a lot in electrical and electronic engineering, e.g., computer clocks, resonant circuits, radio receivers, etc.
- Some could be nonlinear, e.g., Chua's circuit → rich phenomena and more complex, e.g., chaos → not considered here

RLC Circuit as An Oscillator Network

LC circuit as an oscillator



- System dynamics: $C\dot{v}_C = i_C$, $L\dot{i}_L = v_L$.
- Using Kirchhoff's laws: $i_C = i_L$, $v_C + v_L = 0$

$$\Rightarrow \ddot{v}_C = -\frac{1}{LC}v_C$$

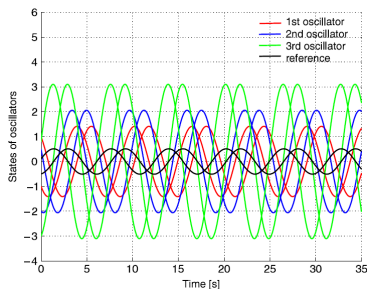
harmonic oscillator

Example Simulation

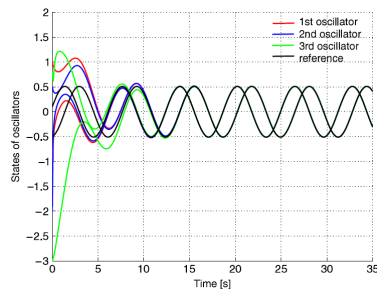
Purpose: harmonic oscillators converge to a given reference oscillator

Example Simulation

Purpose: harmonic oscillators converge to a given reference oscillator



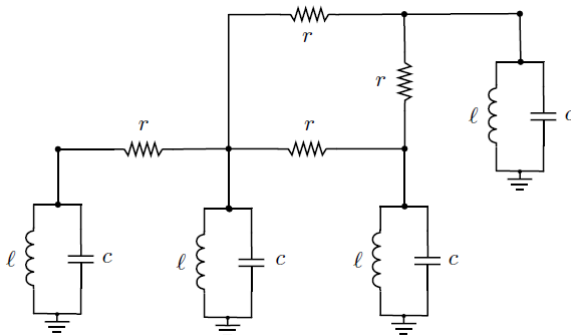
No cooperation



With cooperation

Taken from: Dinh Hoa Nguyen, "A sub-optimal consensus design for multi-agent systems based on hierarchical LQR", Automatica, vol. 55, pp. 88-94, 2015.

RLC Circuit as An Oscillator Network



How about this?